# ALL INDIA INTERNAL TEST SERIES

## CHEMISTRY, MATHEMATICS & PHYSICS

### SET – A

**APT - 3** | **080163.1** | **Ai²TS-6**

**Time Allotted : 3 Hours** **Maximum Marks: 300**

CLASS XII

## INSTRUCTIONS

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

**Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.**

### A. General Instructions

- ✓ Attempt ALL the questions. Answers have to be marked on the OMR sheets.
- ✓ This question paper contains **Three Sections**.
- ✓ **Section – I** is *"Chemistry"*, **Section – II** is *"Mathematics"* and **Section – III** is *"Physics"*.
- ✓ Each Section is further divided into three Parts: **Part – A, Part – B** & **Part – C**.
- ✓ Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
- ✓ Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

(i) **PART-A (01 – 04)** contains 4 Multiple Choice Questions which have Only One Correct answer. Each question carries **+5 marks** for correct answer and **–3 marks** for wrong answer.
**PART-A (05 – 10)** contains 6 Multiple Choice Questions which have One or More Than One Correct answer. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(ii) **PART-B (01 – 04)** contains 4 Matrix Match Type Question which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+8 marks** for all correct answer however for each correct row **+2 marks** will be awarded. There is no negative marking.

(iii) **PART-C (01 – 06)** contains 6 Numerical Based questions with Single Digit Integer as answer, ranging from 0 to 9 and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

**Name of Candidate :**

**Batch ID :** **Date of Examination :** / / 2 0 1

**Enrolment Number :**

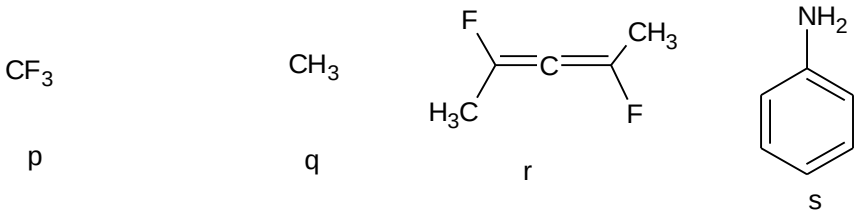
# SECTION- I (CHEMISTRY)

## PART – A

### Straight Objective Type

This section (**01 - 04)** contains 4 multiple choice questions which have only one correct answers. Each question carries **+5 marks** for correct answer and **– 3 mark** for wrong answer.

1.




p q r s

Planar species of above is/are___

(a) p,q (b) p,q,r (c) p,q,r,s (d) q,s

2. 200 ml of 0.15 M $H_3AsO_4$ (given $K_{a_1} = 10^{-3}$ $K_{a_2} = 10^{-7}$ $K_{a_3} = 10^{-12}$); 50 ml of 0.6 M $Na_2HAsO_4$ solutions and 100 ml of 0.1M NaOH are mixed. pH of the final solution is

(a) 6.3 (b) 7.0 (c) 6.7 (d) can not be found.

3. $Mg_2C_3 \xrightarrow{H_2O} Mg(OH)_2 + A$

$A \xrightarrow{red\ hot\ iron\ tube} B$

$A \xrightarrow{H_3O^+/Hg^{2+}} C$

$C \xrightarrow{Conc.\ H_2SO_4} D$

$C(3moles) \xrightarrow{NaOH/\Delta} E$

Then the correct option is....

(a) B and E are same (b) C and D are same

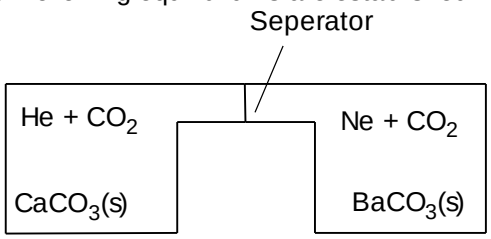
(c) B and D are same (d) D and E are same

4. A container having two partition of equal volume with a separator contains excess of $CaCO_{3(s)}$ in one partition and $BaCO_{3(s)}$ in other partition. First partition contains He and $CO_2$ with total pressure 8 atm and mole fraction of He as 0.50 while second partition contains Ne and $CO_2$ with total pressure of 8 atm and mole fraction of Ne as 0.25. Temperature of the container is 600 K at which following equilibriums are established.

$CaCO_{3(s)} \xrightarrow{600K} CaO_{(s)} + CO_{2(g)}$

$BaCO_{3(s)} \xrightarrow{600K} BaO_{(s)} + CO_{2(g)}$

If the separator is removed then total pressure inside the container will be___(neglect volume change due to solid compounds)




(a) 8 atm (b) 13 atm (c) 9 atm (d) 7 atm

### Straight Objective Type

This section (**05-10)** contains 6 multiple choice questions which have one or more than one correct answers. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

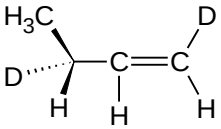
5. If 50 litre flask containing one mole of $N_2$ and one mole of $PCl_5$ is connected to a 50 litre flask containing 2 moles of $PCl_5$, both vessel heated to 227ºC. The equilibrium pressure is found to be 2.05 atm. Assuming ideal behaviour and for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ correct option(s) is/are

(a) degree of dissociation of $PCl_5$ is $\frac{1}{3}$

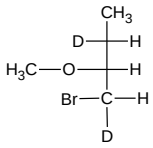
(b) $K_P$ for the reaction is 0.205 atm

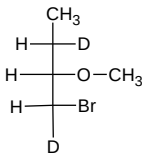
(c) One mole of $PCl_5$ is present in each flask

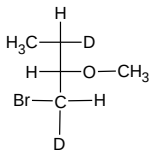
(d) 0.5 moles of all gases are present in each flask.

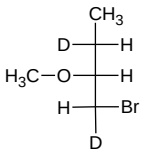
6.  $\xrightarrow{Br_2/CH_3OH}$ P (Mixture of compounds)

P May be :

(a) 

(b) 

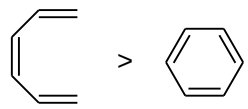
(c) 

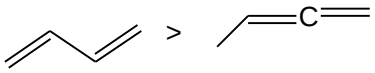
(d) 

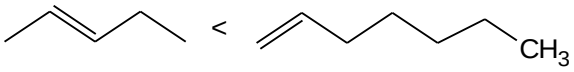
7. Ashutosh collected 1 ltr water in a jar. Kaustub mixed 0.2 mol of $H_2S$ in it while Ananya mixed 0.1 mol of NaOH in it. Arav brought a big piece of NiS and put it in the solution. After long time, some FIITJEE students are observing the solution. Correct observation(s) made by them is/are
( Given $K_{a1}$ ($H_2S$) = $10^{-5}$ $K_{a2}$ ($H_2S$) = $10^{-8}$ $K_{sp}$ (NiS) = $10^{-20}$)

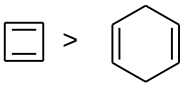
(a) Manish found that Solubility of NiS was increased in that solution from normal solubility (ie s = $10^{-10}$ M) due to reaction of $S^{2-}$ ion with $H^+$ present in the solution.
(b) Nischay found pH of the solution to be 5
(c) Shivang said, "Concentration of [$S^{2-}$] is $10^{-4}$ M"
(d) Utkarsh said, "Concentration of $Ni^{2+}$ is $10^{-10}$ M"

8. Which is ordered correctly

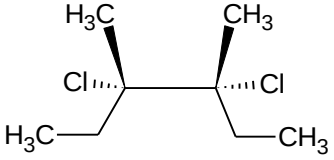
(a)  heat of hydrogenatio

(b)  heat of hydrogenatio

(c)  heat of combustior

(d)  heat of hydrogenatio

9. Consider following reaction and find correct option(s) regarding product(s)

 $\xrightarrow{Cl_2/h\upsilon}$ $C_8H_{15}Cl_3$

All possible compound

(a) Number of optically active compound formed is 10
(b) Number of pair of enantiomers is 5
(c) Number of pair of diastereomers formed is 4
(d) Number of optically inactive compounds formed is 2

10. Consider following equilibriums :

$A(s) \underset{}{\overset{K_{p_1}}{\rightleftharpoons}} B(g) + C(g)$

$E(s) \underset{}{\overset{K_{p_2}}{\rightleftharpoons}} B(g) + D(g)$

$P(s) \underset{}{\overset{K_{p_3}}{\rightleftharpoons}} B(g) + Q(s)$

Pressure above the solids $A_{(s)}$ , $E_{(s)}$ and $P_{(s)}$ when kept alone are 4 atm, $8\sqrt{2}$ atm and 5 atm respectively at temperature T(K). When three solids A(s),s E(s), P(s) are kept together at the same temperature T(K) then.
(a) Partial pressure of $B_{(g)}$ is 5 atm (b) Partial pressure of $C_{(g)}$ is 4/5 atm
(c) Partial pressure of $D_{(g)}$ is 32/5 atm (d) Total pressure is 12 atm.

## SECTION – B

### Matrix Match Type

This section **(01 – 04)** contains 4 Matrix Match Type Question which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+8 marks** for all correct answer however for each correct row **+2 marks** will be awarded. There is no negative marking.

1. Match the Column-I with Column – II (where $\alpha$ = degree of dissociation, $M_{mix}$ = average molecular weight of mixture ; P = equilibrium pressure)

| COLUMN – I | | COLUMN – II | |
|---|---|---|---|
| (a) | $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}\ \Delta H = -ve$ | (p) | $\alpha = \sqrt{\frac{K_p}{4P + K_p}}$ |
| (b) | $D_{(g)} \rightleftharpoons 2E_{(g)} + Energy$ | (q) | Yield of product(s) will be high at low temperature |
| (c) | $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ | (r) | Yield of product(s) will be high at high temperature |
| (d) | $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ | (s) | $M_{mix} = \frac{M_{reactant}}{1+\alpha}$ |
| | | (t) | Yield of product(s) will be high at low pressure |

2.

| Column – I (Reactions) | | Column – II (Product(s)) | |
|---|---|---|---|
| (a) | Borax $\xrightarrow{\Delta}$ | P | $H_2B_4O_7$ |
| (b) | Borax + $H_2O$ $\longrightarrow$ | Q | $H_3BO_3$ |
| (c) | Borax + NaOH $\longrightarrow$ | R | Used in glass formation |
| (d) | Borax + HCl $\longrightarrow$ | S | $NaBO_2$ |
| | | T | $B_2O_3$ |

3. Match the Column-I with Column – II

| Column – I (Mixing equal volume for $H_2S$ $Ka_1 = 10^{-6}$ $Ka_2 = 10^{-9}$) | | Column – II | |
|---|---|---|---|
| (a) | 1 M $H_2S$ + 1 M $Na_2S$ | (P) | pH =7.5 |
| (b) | 2 M $H_2S$ + 1 M $Na_2S$ | (Q) | pH = 6 |
| (c) | 1 M $H_2S$ + 2 M $Na_2S$ | (R) | pH = 9 |
| (d) | 1 M $Na_2S$ + 2 M HCl | (S) | Buffer |
| | | (T) | pH = 3.15 |

4.

| COLUMN – I | | COLUMN – II | |
|---|---|---|---|
| (a) | cis–2-butene $\xrightarrow{Br_2/CCl_4}$ | (p) | Cyclic intermediate |
| (b) | Trans-2-butene $\xrightarrow[alk.KMnO_4]{cold}$ | (q) | Recemic mixture |
| (c) | Trans-2-butene $\xrightarrow{Br_2/CS_2}$ | (r) | Meso-isomer |
| (d) | Cis-2-butene $\xrightarrow{HOCl}$ | (s) | Enantiomeric pairs |
| | | (t) | Product/s mixture is/are optically inactive |

## PART – C

### (Integer Type)

This section contains **6 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

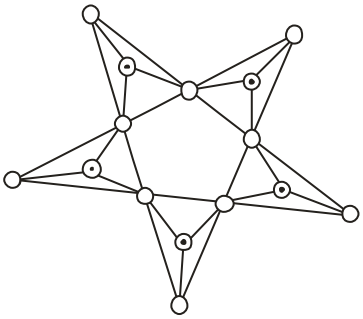
1. Maximum pH that must be maintained in a saturated $H_2S$ (0.1 M) to precipitate CdS but not ZnS is x and concentration of $Cd^{2+}$ ion at that pH is $y \times 10^{-7}$ then x + y =....
given $[Cd^{2+}] = [Zn^{2+}] = 0.1$ M initially ?

$K_{sp}$ (CdS) = $8 \times 10^{-27}$

$K_{sp}$ (ZnS) = $1 \times 10^{-21}$

$K_{a1}$ ($H_2S$) = $1.0 \times 10^{-8}$ ,$K_{a2}$ ($H_2S$) = $1.0 \times 10^{-13}$

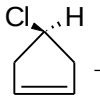
2.



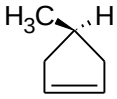
One molecule of silicate mineral containing above anion has x number of atoms of a metal M, y number of atoms of Potassium and z number of atoms of Calcium as cations.

If $x^2+y^2+2z^2 = 2yz + 2x + 4z - 5$ then oxidation state of metal M is___

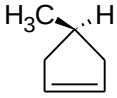
3.




$\xrightarrow{HCl}$ P (Number of Products)




$\xrightarrow{ICl/CCl_4}$ Q (Number of Products)




$\xrightarrow{\text{Cold Dil Alk } KMnO_4}$ R (Number of Products)

Value of P + Q + R is _____

4. The equilibrium composition in 1 litre container for the reaction is -

$PCl_3 + Cl_2 \rightleftharpoons PCl_5$

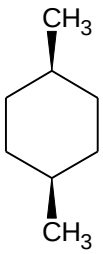
0.20 0.08 0.40 moles/litre

0.22 mole of $Cl_2$ is added at same temperature and new equilibrium is established and If sum of new equilibrium concentration of $PCl_3$, $Cl_2$ and $PCl_5$ is $p$ then value of $p \times 10$ is_____

5. $B_2H_6 + NH_3 \xrightarrow[\text{low temperature}]{\text{excess of } NH_3} P \xrightarrow{\text{strong heating}} Q(\text{an aromatic compound})$

Number of bond between B and N in cation part of P is x, in anion part of P is y and in compound Q is z then the sum x+y+z =….(count only sigma or coordinate bond)

6. Number of diastereomer pairs that can be grouped from the products formed by mono chlorination of following compound is...




$\xrightarrow{Cl2/h\nu}$ *products*

# SECTION- II (MATHMETICS)

## PART – A

### Straight Objective Type

This section (**01 - 04)** contains 4 multiple choice questions which have only one correct answers. Each question carries **+5 marks** for correct answer and **– 3 mark** for wrong answer.

1. The normal chord to the parabola $y^2 = 8x$ at the point (8, 8) subtends an angle $\theta$ at the focus, where $\theta$ is :

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$

2. The locus of the point of intersection of tangents to an ellipse at two points, sum of whose eccentric angle is constant is a part of a/an :

(a) ellipse (b) hyperbola (c) parabola (d) straight line

3. Equation of a line passing through the centre of a rectangular hyperbola is x – y – 1 = 0. If one if of its asymptotes is 3x – 4y – 6 = 0 and the equation of other asymptote is ax + by + c = 0, (where a, b, c $\in N$), then the minimum value of a + b + c is

(a) 10 (b) 24 (c) 22 (d) 28

4. If the slope of one of the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$ be k times that of the other, then :

(a) $4kh = ab(1 + k)$ (b) $kh = ab(1 + k)^2$ (c) $4kh^2 = ab(1 + k)^2$ (d) None of these

### Straight Objective Type

This section (**05-10)** contains 6 multiple choice questions which have one or more than one correct answers. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

5. If y = 3x + 1 is a tangent to the hyperbola $\frac{x^2}{p^2} - \frac{y^2}{8} = 1$, then which of the following cannot be the sides of a right angle triangle ?

(a) $p^4, \sqrt{10}, 3$ (b) $p^4, 10, \sqrt{3}$ (c) $\sqrt{p^4}, 10, 3$ (d) $\sqrt{p^4}, \sqrt{10}, 3$

6. If $y\sin(x)\left(\frac{dy}{dx}\right) = \cos(x)[\sin(x) - y^2]$ and $y = \sqrt{\frac{2}{3}}$ at $x = \frac{\pi}{2}$, then the value of y at $x = \frac{\pi}{3}$ is NOT equal to

(a) $3^{1/2}$ (b) $3^{1/4}$ (c) $3^{-1/4}$ (d) $3^{-1/2}$

7. If the length of intercepts made on the line passing through origin and x + y = 1 by the circle $x^2 + y^2 = x - 3y$ are equal, then the above mentioned line can be

(a) x + y = 0 (b) x – y = 0 (c) x + 7y = 0 (d) x – 7y = 0

8. If the line y = 2x + a lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, then a can be

(a) – 2 (b) – 4 (c) – 6 (d) – 8

9. If the line x = P divides the area of the region $S = \{(x,y) \in R \times R;\ x^3 \le y \le x, 0 \le x \le 1\}$ into two equal parts, then

(a) $0 < P \le \frac{1}{2}$ (b) $\frac{1}{2} < P < 1$ (c) $2P^4 - 4P^2 + 1 = 0$ (d) $P^4 + 4P^2 - 1 = 0$

10. If the curve passing through (0, 1) and satisfying the differential equation $\frac{dy}{dx} = \frac{y}{(2y\ell n(y) + y - x)}$ is y = f(x), then f(e) is greater than

(a) e (b) 1 (c) $\pi$ (d) 2

## SECTION – B

### Matrix Match Type

This section **(01 – 04)** contains 4 Matrix Match Type Question which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+8 marks** for all correct answer however for each correct row **+2 marks** will be awarded. There is no negative marking.

1. Match the following :

| **Column – I** | | **Column – II** | |
|---|---|---|---|
| (a) | 4y = 5x is a secant to the curve(s) | (P) | $x^2 + y^2 = 1$ |
| (b) | The curve passing through (1, 0) and satisfying $[1+y^2] = xy\left(\frac{dy}{dx}\right)$ is/are | (Q) | $x^2 - y^2 = 1$ |
| (c) | The curve for which the lines $x^2 = 1$ are tangent is/are | (R) | $2x^2 - y^2 = 2$ |
| (d) | $\left(0, \frac{5}{4}\right)$ is an exterior point for | (S) | $2x^2 + y^2 = 2$ |

2. For the parabola $(x-2)^2 + (y-3)^2 = \frac{(3x-y+7)^2}{10}$

| **Column – I** | | **Column – II** | |
|---|---|---|---|
| (a) | The equation of line about which parabola is symmetrical | (P) | 3x – y + 7 = 0 |
| (b) | The equation of line along which minimum length of focal chord occurs | (Q) | 3x – y – 3 = 0 |
| (c) | The locus of point of intersection of perpendicular tangents | (R) | x + 3y – 11 = 0 |
| (d) | The locus of foot of perpendicular from focus upon any tangent | (S) | 3x – y + 2 = 0 |

3. Match the following

| **Column – I** | | **Column – II** | |
|---|---|---|---|
| (a) | If the circles $x^2 + y^2 = b^2$ and $x^2 + y^2 + 2gx + 2fy + ac = 0$ cut each other orthogonally, then | (P) | $ac \geq 0$ |
| (b) | If the circles $x^2 + y^2 + 2ax + b = 0$ and $x^2 + y^2 + 2cx + b = 0$ touch each other $(a \neq c)$ | (Q) | $b^2 = ac$ |
| (c) | If in (b), the first circle lies completely inside the second circle, then | (R) | b > 0 |
| (d) | If the chord of contact of the tangents drawn to $x^2 + y^2 = b^2$ from any point on $x^2 + y^2 = a^2$, touches the circle $x^2 + y^2 = c^2$ then | (S) | b = 0 |
| | | (T) | $\lvert b \rvert \leq 0$ |

4. Match the following

| **Column – I** | | **Column – II** | |
|---|---|---|---|
| (a) | The area bounded by $x^2 \leq y$ and $y \leq x + 2$ is | (P) | -12 |
| (b) | The sum of the distance of any point on the ellipse $3x^2 + 4y^2 = 12$ from its directrix is | (Q) | 9/2 |
| (c) | The value of c for which $3x^2 - 5xy - 2y^2 + 5x + 11y + c = 0$ are the asymptotes of the hyperbola $3x^2 - 5xy - 2y^2 + 5x + 11y - 8 = 0$ is | (R) | 8 |
| (d) | If e is the eccentricity of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\theta$ is the angle between the asymptotes then $8e\cos\left(\frac{\theta}{2}\right)$ equals | (S) | $\frac{\sqrt{386}}{88}$ |

## PART – C
## (Integer Type)

This section contains **6 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

1. A curve y = f(x) passes through origin and slope of tangent line at any point (x, y) of the curve is $\frac{x^4+2xy-1}{1+x^2}$, then the value of greatest integer which is less than or equal to f(-1) is

2. If the equation of the directrix of the parabola which touches x-axis and y-axis at (2, 0) and (0, 3) respectively, is ax + by + c = 0, (a, b, c are integers) then the minimum value of |a + b + c| is

3. If the differential equation corresponding to the family of curves $y = (A + Bx)e^{3x}$ is given by $\frac{d^2y}{dx^2} = a\frac{dy}{dx} + by$, then $\frac{(a-b)}{5}$ equals

4. Let A be a point on $\frac{(x-2)^2}{16} + \frac{y^2}{12} = 1$, B and C be its foci. If the locus of the incentre of $\Delta ABC$ is an ellipse with eccentricity e, then $9e^2$ is equal to

5. If (x, y) satisfies the equation $x^2 + y^2 – 6x – 8y + 21 = 0$ then the minimum value of $\frac{1}{2}(x^2+y^2+1)+(x-y)$ is equal to

6. Chords of the hyperbola $x^2 – y^2 = a^2$ touch the parabola $y^2 = 4ax$. If the locus of their middle points is the curve $y^m(x - a) = x^n$, then m +n equals

# SECTION- III (PHYSICS)

## PART – A

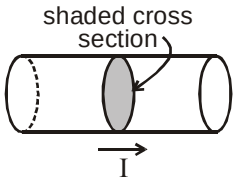
### Straight Objective Type

This section (**01 - 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+5 marks** for correct answer and **– 3 mark** for wrong answer.

1. A variable resistance is connected across a non ideal cell of constant emf and constant internal resistance. Initial value of variable resistance is R. Now the magnitude of variable resistance has been made n times, that is, its new value is nR. Surprisingly the new value of power dissipated in variable resistance is still same as its initial value. The value of internal resistance of cell is

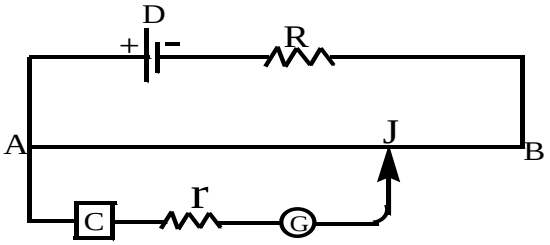
(A) $\frac{1+n}{2}R$ (B) $R\sqrt{n}$ (C) $\sqrt{\frac{1+n^2}{2}}R$ (D) $R+nR$

2. A current I flows through a cylindrical rod of uniform cross-section area A and resistivity $\rho$ . The electric flux through the shaded cross-section of rod as shown in figure is :




(A) $\frac{I}{\rho}$ (B) $\rho I$ (C) $\frac{\rho I}{A}$ (D) $\frac{\rho A}{I}$

3. In the given potentiometer circuit, the resistance of the potentiometer wire AB is $R_0$ . 'C'is a cell whose internal resistance r is separately shown. The zero centered galvanometer '*G*' does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this?



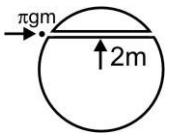

(A) r > $R_0$

(B) R >> $R_0$

(C) emf of cell C > emf of cell D

(D) The negative terminal of C is connected to A.

4. A unit positive point charge of mass $\pi$ gm is to be projected inside the tunnel in sphere of radius 4m (as shown in figure), with a velocity of 5 m/sec. The tunnel has been made inside a uniformly charged $(\rho = 1\,c/m^3)$ non conducting sphere. The minimum velocity with which the point charge should be projected such that it can reach the opposite end of the tunnel, is equal to

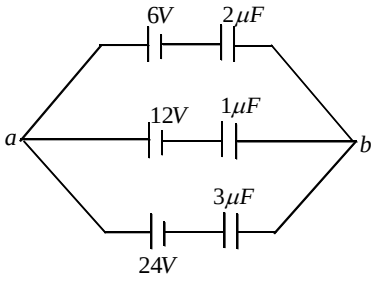



(a) $1.0 \times 10^6$ m/sec
(b) $12 \times 10^6$ m/sec
(c) zero, because the initial and the final points are at same potential
(d) $4.8 \times 10^6$ m/sec

**Straight Objective Type**

This section (**05-10)** contains 6 multiple choice questions which have one or more than one correct answers. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

5. In the given electric circuit which of the following statements are correct (Assume that the given capacitors are uncharged before they are connected in the circuit and the batteries of negligible internal resistance)




A) $V_a - V_b = -16V$ B) Charge on the $2\mu F$ capacitor is $20\mu C$

C) Charge on the $1\mu F$ capacitor is $4\mu C$ D) Charge on the $3\mu F$ capacitor is $24\mu C$

6. A ring rolls without slipping on the ground. Its centre C moves with a constant sped u. P is any point on the ring. The speed of P with respect to the ground is v, then

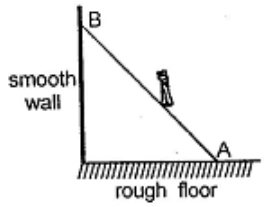
A) $0 \le v \le 2u$

B) v = u, if CP is horizontal

C) $v = \sqrt{6}u$ , if CP makes an angle of $30^0$ with the horizontal

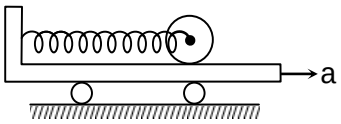
D) $v = \sqrt{2}u$ , if CP is horizontal

7. A ladder AB is supported by a smooth vertical wall and rough horizontal floor as shown. A boy starts moving from A to B slowly. The ladder remains at rest, then pick up the correct statements(s):



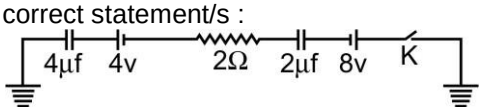

a) Magnitude of normal reaction by wall on ladder at point B will increase.
b) Magnitude of normal reaction by wall on ladder at point B will decrease.
c) Magnitude of normal reaction by floor on ladder at point A will remain unchanged
d) Magnitude of friction force by floor on ladder at point A will increase

8. A solid uniform sphere is connected with a moving trolley car by a light spring. The trolley car moves with an acceleration a. If the sphere remains at rest relative to the trolley car, then :




(A) Spring force = ma
(B) Friction between the sphere and trolley car is equal to zero
(C) Friction between the sphere and trolley car is equal to $\frac{ma}{2}$
(D) Spring force is equal to $\frac{ma}{2}$

9. As shown in the figure find the correct statement/s :




(a) i at instant switch is closed is 2A
(b) i after long time is 2A
(c) charge on 4μf after long time is $\frac{16}{3}\mu c$ (when switch is closed)

(d) charge on $2\mu f$ at instant switch is closed is $8\mu c$

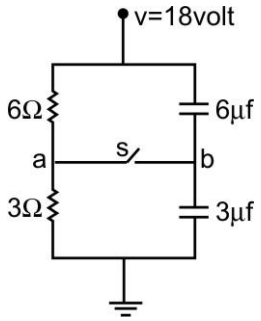
10. Initially switch is open. After some time switch is closed. Choose the correct statements

(a) the charge flown through the switch is 54 $\mu c$

(b) Point b is at higher potential than point a before the switch is closed

(c) the potential of point b is 6v w.r.t. ground, after s is closed

(d) the potential of point b is 12v, after s is closed

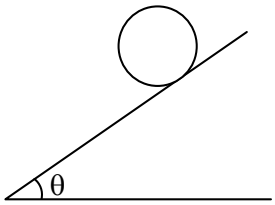



## SECTION – B

### Matrix Match Type

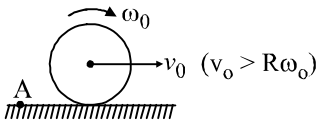
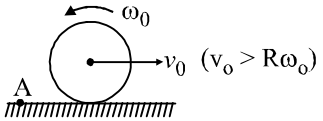
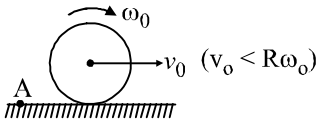
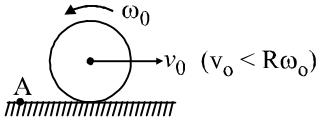
This section **(01 – 04)** contains 4 Matrix Match Type Question which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+8 marks** for all correct answer however for each correct row **+2 marks** will be awarded. There is no negative marking.

1. A ring of mass m and radius R is placed on a rough inclined plane so that it rolls without slipping. Match the following table.




| | Column I | | Column II |
|---|---|---|---|
| (A) | Linear acceleration of centre of mass | (p) | Is directly proportional to m |
| (B) | Angular acceleration | (q) | Is inversely proportional to m |
| (C) | Rotational kinetic energy at any instant | (r) | Is directly proportional to $R^2$ |
| (D) | Translational kinetic energy at any instant | (s) | Is directly proportional to $R^3$ |
| | | (t) | None of these |

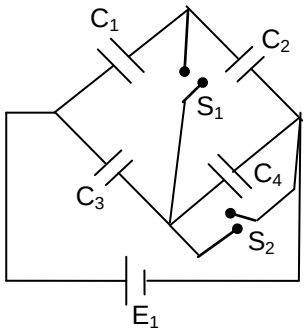
**2.** In each situation of column–I a uniform disc of mass m and radius R rolls on a rough fixed horizontal surface as shown. At t = 0 (initially) the angular velocity of disc is $\omega_o$ and velocity of centre of mass of disc is $v_o$ (in horizontal direction). The relation between $v_o$ and $\omega_o$ for each situation and also initial sense of rotation is given for each situation in column–I. Then match the statements in column–I with the corresponding results in column–II

| Column-I | | Column-II | |
|---|---|---|---|
| (A) |   | (P) | The angular momentum of disc about point A (as shown in figure) remains conserved. |
| (B) |   | (Q) | The kinetic energy of disc after it starts rolling without slipping is less than its initial kinetic energy |
| (C) |   | (R) | In the duration disc rolls with slipping, the friction force on disc acts backward. |
| (D) |   | (S) | Before rolling starts acceleration of the disc remain constant in magnitude and direction |
| | | (T) | Final angular velocity is independent friction coefficient between disc and the surface |

3. A galvanometer has a resistance of $100\Omega$ and a full scale range of $100\mu A$. It can be used as a higher range ammeter or a voltmeter by providing a resistance. Match the statements of the two columns :

| Column - I | | Column - II | |
|---|---|---|---|
| (a) | 25 V range with | (P) | $200K\Omega$ resistance in series |
| (b) | 20 V range with | (Q) | $10K\Omega$ resistance in series |
| (c) | 5 mA range with | (R) | $1\Omega$ resistance in parallel |
| (d) | 10 mA range with | (S) | None of these |
| | | (T) | $2\ \Omega$ resistance in parallel |

4. Four capacitors with capacitances $C_1 = 2\mu F$, $C_2 = 4\mu F$, $C_3 = 4\mu F$, $C_4 = 2\mu F$ are connected as shown in figure and are connected to a 12 V source. Charge flow battery $\Delta q_E$ and switch is $\Delta q_S$.



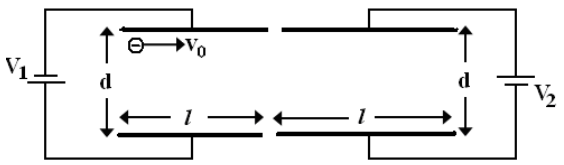

| List I | | | List II | |
|---|---|---|---|---|
| (A) | Switch $S_1$ is closed value of $\Delta q_E$ | | (P) | 32 $\mu$C |
| (B) | Switch $S_1$ is closed value of $\Delta q_S$ | | (Q) | 4 $\mu$C |
| (C) | Switch $S_2$ is closed value of $\Delta q_E$ | | (R) | 48 $\mu$C |
| (D) | Switch $S_1$ is closed value of $\Delta q_S$ | | (S) | 12 $\mu$C |

## PART – C
## (Integer Type)

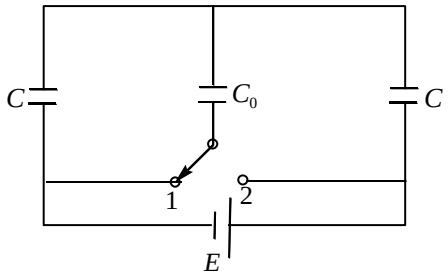
This section contains **6 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

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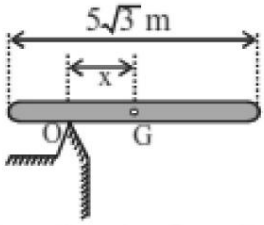
1. Two identical parallel plate capacitors are placed side by side with a small gap between them as shown in fig. $V_1$ and $V_2$ are potential difference across the two capacitors respectively. An electron is projected from the end of upper plate of the left capacitor with velocity $v_0$ towards right as shown. Charge and mass of electron 'e' and 'm' respectively. $V_2$ is applied such that the electron just grazes the lower plate of right capacitor horizontally while coming out of it. Assume that there is no gravity and no collision of electron with any plate. What is the ratio $\frac{V_2}{V_1}$? (neglect the time taken by electron from one capacitor to another capacitor.)



2. What amount of heat in micro joule will be generated in the circuit , after the switch is shifted from position 1 to position 2 ? $\left(C = C_0 = 2\mu F, E = \sqrt{6}\,V\right)$

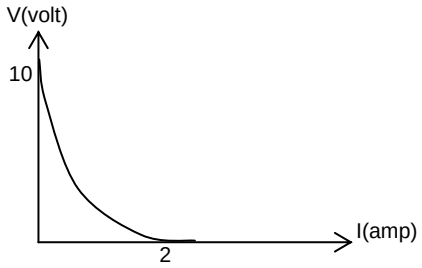



3.

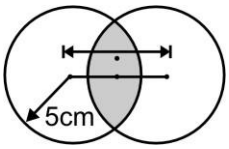



In the adjacent figure a uniform rod of length $5\sqrt{3}m$ and mass 2 kg is kept at rest in horizontal position on an elevated edge. Friction is sufficient to prevent slipping between rod and elevated edge. The value of x (consider the figure) is chosen such that the rod will have maximum angular acceleration $\alpha$, as soon as it is set free. Find the maximum angular acceleration of the rod (in rad / $s^2$).

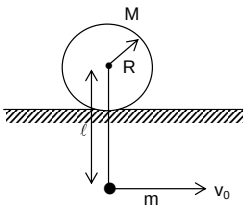
4. A battery of emf E and internal resistance r is connected with external resistance R. Resistance R can be adjusted to any value greater than or equal to zero. Graph is plotted between the current passing through the resistance (I) and potential difference (V) across it. Internal resistance of the battery ( in $\Omega$ ) is




5. Two identical intersecting spheres having the same volume charge density of magnitude $\varepsilon_0$ (in S.I. unit) opposite in nature of charge, are shown in figure. Radius of spheres are 5cm. The distance between the centres is 6cm. Then the magnitude of electric field in the shaded portion is $\frac{E_0}{100}$, then $E_0$ will to (In S.I. unit)




6. A disc of radius R and mass M is placed on smooth horizontal surface as shown in figure. A light rod of length $\ell$ is hanged from the centre of disc and a small mass m is attached at the end as shown in figure. Now a velocity $V_0$ is given to mass m. The maximum height which mass m can attain. $[v_0 = \sqrt{8g}, M = 3m]$




( Assume rod can rotate < 90°)